



Synthetic Division

As long as you're careful, long division always works, but it takes time. Another method you can use to divide polynomials is synthetic division.

For synthetic division to work, the divisor must be a binomial in the form $x - k$, where k is a constant.

Using Synthetic Division to Divide Polynomials

Suppose you want to divide $2x^2 + 3x + 7$ by $x - 2$. The process of finding the quotient using both long division and synthetic division is shown.

Long Division

$$\begin{array}{r} \overline{2x+7} \\ x-2 \overline{) 2x^2 + 3x + 7} \\ \underline{-(2x^2 - 4x)} \\ 7x + 7 \\ \underline{-(7x - 14)} \\ 21 \end{array}$$

Synthetic Division

$$\begin{array}{r|rrrr} 2 & & 2 & 3 & 7 \\ & & & 4 & 14 \\ \hline & & 2 & 7 & 21 \end{array}$$

Synthetic division is considered a shorthand method for long division, but remember that it only works when the divisor is in the form $x - k$.

► **Think About It** When you use synthetic division, notice that the coefficients of the dividend appear in the top row and the coefficients of the quotient appear with the remainder in the bottom row.

EXAMPLE 1

Divide $3x^3 + 11x^2 - 2x - 24$ by $x + 3$.

SOLUTION

The divisor is $x + 3$. To get the form $x - k$, write $x + 3$ as $x - (-3)$, so $k = -3$.

Step 1

$$\begin{array}{r|rrrr} -3 & 3 & 11 & -2 & -24 \\ \hline \end{array}$$

Put the value for k in the box and put the coefficients to the right of it.

Step 2

$$\begin{array}{r|rrrr} -3 & 3 & 11 & -2 & -24 \\ & & -9 & & \\ \hline & 3 & & & \end{array}$$

Bring down the first coefficient, 3, and then multiply by -3 . Put the result in the next column.

Step 3

$$\begin{array}{r|rrrr} -3 & 3 & 11 & -2 & -24 \\ & & -9 & -6 & \\ \hline & 3 & 2 & & \end{array}$$

Add 11 and -9 and then multiply the sum, 2, by -3 . Put the result in the next column.

Step 4

$$\begin{array}{r|rrrr} -3 & 3 & 11 & -2 & -24 \\ & & -9 & -6 & 24 \\ \hline & 3 & 2 & -8 & \end{array}$$

Add -2 and -6 and then multiply the sum, -8 , by -3 . Put the product in the next column.

Step 5

$$\begin{array}{r|rrrr} -3 & 3 & 11 & -2 & -24 \\ & & -9 & -6 & 24 \\ \hline & 3 & 2 & -8 & 0 \end{array}$$

Add -24 and 24. The sum, 0, is the remainder.

Step 6

$$\begin{array}{r}
 \begin{array}{r|rrrr}
 -3 & 3 & 11 & -2 & -24 \\
 & & -9 & -6 & 24 \\
 \hline
 & 3 & 2 & -8 & 0 \\
 & \downarrow & \downarrow & \downarrow & \downarrow \\
 & 3x^2 & +2x & -8 & R0
 \end{array}
 \end{array}$$

Write the quotient. Begin writing the quotient with a power of x that is one less than the greatest power of x in the dividend.

The quotient has no remainder, so

$$(3x^3 + 11x^2 - 2x - 24) \div (x + 3) = 3x^2 + 2x - 8. \blacksquare$$

EXAMPLE 2

Divide $x^3 - 6x^2 - x + 32$ by $x - 5$.

SOLUTION

The divisor is $x - 5$, so $k = 5$.

Step 1

$$\begin{array}{r|rrrr}
 5 & 1 & -6 & -1 & 32 \\
 \hline
 \end{array}$$

Put the value for k in the box and put the coefficients to the right of it.

Step 2

$$\begin{array}{r|rrrr}
 5 & 1 & -6 & -1 & 32 \\
 & & 5 & & \\
 \hline
 & 1 & & &
 \end{array}$$

Bring down the first coefficient, 1, and then multiply by 5. Put the result in the next column.

Step 3

$$\begin{array}{r|rrrr}
 5 & 1 & -6 & -1 & 32 \\
 & & 5 & -5 & \\
 \hline
 & 1 & -1 & &
 \end{array}$$

Add -6 and 5 and then multiply the sum, -1 , by 5 . Put the result in the next column.

Step 4

$$\begin{array}{r|rrrr}
 5 & 1 & -6 & -1 & 32 \\
 & & 5 & -5 & -30 \\
 \hline
 & 1 & -1 & -6 &
 \end{array}$$

Add -1 and -5 and then multiply the sum, -6 , by 5 . Put the product in the next column.

Step 5

$$\begin{array}{r|rrrr}
 5 & 1 & -6 & -1 & 32 \\
 & & 5 & -5 & -30 \\
 \hline
 & 1 & -1 & -6 & 2
 \end{array}$$

Add 32 and -30 . The sum, 2 , is the remainder.

Step 6

$$\begin{array}{r|rrrr}
 5 & 1 & -6 & -1 & 32 \\
 & & 5 & -5 & -30 \\
 \hline
 & 1 & -1 & -6 & 2 \\
 & \downarrow & \downarrow & \downarrow & \downarrow \\
 & x^2 - x & - 6 & & R2
 \end{array}$$

Write the quotient. Begin writing the quotient with a power of x that is one less than the greatest power of x in the dividend.

The quotient has remainder 2 , so

$$(x^3 - 6x^2 - x + 32) \div (x - 5) = x^2 - x - 6 + \frac{2}{x - 5} \cdot \blacksquare$$

EXAMPLE 3

Divide $2x^4 + 9x^3 - 7x + 36$ by $x + 4$.

SOLUTION

The divisor is $x + 4$. To get the form $x - k$, write $x + 4 = x - (-4)$ so $k = -4$.

The dividend is missing an x^2 term, so use 0 for that coefficient.

Step 1

$$\begin{array}{r|rrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 \hline
 \end{array}$$

Put the value for k in the box and put the coefficients to the right of it.

Step 2

$$\begin{array}{r|rrrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 & & -8 & & & \\
 \hline
 & 2 & & & &
 \end{array}$$

Bring down the first coefficient, 2, and then multiply by -4 . Put the result in the next column.

Step 3

$$\begin{array}{r|rrrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 & & -8 & -4 & & \\
 \hline
 & 2 & 1 & & &
 \end{array}$$

Add 9 and -8 and then multiply the sum, 1, by -4 . Put the result in the next column.

Step 4

$$\begin{array}{r|rrrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 & & -8 & -4 & 16 & \\
 \hline
 & 2 & 1 & -4 & &
 \end{array}$$

Add 0 and -4 and then multiply the sum, -4 , by -4 . Put the product in the next column.

Step 5

$$\begin{array}{r|rrrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 & & -8 & -4 & 16 & -36 \\
 \hline
 & 2 & 1 & -4 & 9 &
 \end{array}$$

Add -7 and 16 and then multiply the sum, 9, by -4 . Put the product in the next column.

Step 6

$$\begin{array}{r|rrrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 & & -8 & -4 & 16 & -36 \\
 \hline
 & 2 & 1 & -4 & 9 & 0
 \end{array}$$

Add 36 and -36 . The sum, 0, is the remainder.

Step 7

$$\begin{array}{r|rrrrrr}
 -4 & 2 & 9 & 0 & -7 & 36 \\
 & & -8 & -4 & 16 & -36 \\
 \hline
 & 2 & 1 & -4 & 9 & 0 \\
 & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 & 2x^3 & + x^2 & - 4x & + 9 & R0
 \end{array}$$

Write the quotient. Begin writing the quotient with a power of x that is one less than the greatest power of x in the dividend.

The quotient has no remainder, so

$$(2x^4 + 9x^3 - 7x + 36) \div (x + 4) = 2x^3 + x^2 - 4x + 9. \blacksquare$$

Dividing Polynomials Using Synthetic Division When the Divisor Is Not in the Form $x - k$

EXAMPLE 4

Divide $4y^3 - 5y - 10$ by $2y - 3$.

SOLUTION

To use synthetic division, you must have a divisor in the form $x - k$. Rewrite the division problem by factoring 2 from the divisor and the dividend.

$$\frac{4y^3 - 5y - 10}{2y - 3} = \frac{2\left(2y^3 - \frac{5}{2}y - 5\right)}{2\left(y - \frac{3}{2}\right)} = \frac{2y^3 - \frac{5}{2}y - 5}{y - \frac{3}{2}}$$

The dividend is now $2y^3 - \frac{5}{2}y - 5$ and the divisor is $y - \frac{3}{2}$, so $k = \frac{3}{2}$.

The dividend is missing a y^2 term, so use 0 for that coefficient.

Step 1

$$\begin{array}{r|rrrrr} \frac{3}{2} & & 2 & 0 & -\frac{5}{2} & -5 \\ \hline \end{array}$$

Put the value for k in the box and put the coefficients to the right of it.

Step 2

$$\begin{array}{r|rrrrr} \frac{3}{2} & \textcolor{red}{2} & 0 & -\frac{5}{2} & -5 & \\ \hline & & 3 & & & \\ \textcolor{red}{2} & & & & & \end{array}$$

Bring down the first coefficient, 2, and then multiply by $\frac{3}{2}$. Put the result in the next column.

Step 3

$$\begin{array}{r|rrrrr} \frac{3}{2} & & 2 & \textcolor{red}{0} & -\frac{5}{2} & -5 \\ \hline & & & \textcolor{red}{3} & \frac{9}{2} & \\ & 2 & \textcolor{red}{3} & & & \end{array}$$

Add 0 and 3 and then multiply the sum, 3, by $\frac{3}{2}$. Put the result in the next column.

Step 4

$$\begin{array}{r}
 \frac{3}{2} \overline{) \begin{array}{cccc} 2 & 0 & -\frac{5}{2} & -5 \\ & 3 & \frac{9}{2} & 3 \\ \hline 2 & 3 & 2 & \end{array}}
 \end{array}$$

Add $-\frac{5}{2}$ and $\frac{9}{2}$ and then multiply the sum, 2, by $\frac{3}{2}$. Put the product in the next column.

Step 5

$$\begin{array}{r}
 \frac{3}{2} \overline{) \begin{array}{cccc} 2 & 0 & -\frac{5}{2} & -5 \\ & 3 & \frac{9}{2} & 3 \\ \hline 2 & 3 & 2 & -2 \end{array}}
 \end{array}$$

Add -5 and 3 . The sum, -2 , is the remainder.

Step 6

$$\begin{array}{r}
 \frac{3}{2} \overline{) \begin{array}{cccc} 2 & 0 & -\frac{5}{2} & -5 \\ & 3 & \frac{9}{2} & 3 \\ \hline 2 & 3 & 2 & -2 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2y^2 + 3y & + 2 & R-2 \end{array}}
 \end{array}$$

Write the quotient. Begin writing the quotient with a power of x that is one less than the greatest power of x in the dividend.

The quotient has a remainder of -2 , so

$$\begin{aligned}
 (4y^3 - 5y - 10) \div (2y - 3) &= 2y^2 + 3y + 2 - \frac{2}{y - \frac{3}{2}} \\
 &= 2y^2 + 3y + 2 - \frac{4}{2y - 3}. \quad \blacksquare
 \end{aligned}$$

► **Remember** To simplify a compound fraction, multiply the numerator and denominator by their least common denominator.

$$\frac{2}{y - \frac{3}{2}} = \frac{2 \cdot 2}{2 \cdot \left(y - \frac{3}{2}\right)} = \frac{4}{2y - 3}$$